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Artificial Intelligence **Applications** · ITAI 2372

AI that doesn't just predict. AI that proves.



Machine Learning



Data Pipelines



AI Systems



Smart Infrastructure



Applied AI Ethics

AGI in Healthcare

The Physician in the Machine — from Narrow AI scribes to a cross-modal Super-Consultant



WHAT I LEARNED

Today's medical AI is good at one task — spotting a tumor on a scan, for example. I explored what a smarter version would look like: a system that extracts data from a patient's wearable data, DNA, and environment to understand the whole person. Then I weighed the real risks — AI that can't explain its decisions, bias that spreads across millions of patients, and doctors who lose their edge when they stop thinking for themselves.

CONCEPTS MASTERED

- Narrow AI vs AGI: cross-domain reasoning
- Cross-modal synthesis & digital-twin simulation
- Black-box risk, bias scaling, de-skilling

N A R R O W A I	→	A G I
Reasoning	<i>Single-task</i>	Cross-domain
Diagnosis	<i>Reads a scan</i>	Reads the patient
Drug response	<i>Trial & error</i>	Digital-twin sim
Access	<i>Top hospitals</i>	Anywhere on Earth
Care model	<i>One size fits all</i>	Tuned to your DNA

AI in Manufacturing

GE predictive maintenance + proposed AI-driven energy optimization



WHAT I LEARNED

Examined how General Electric's Predix platform combined IoT sensors and machine learning to predict equipment failures before they happened, then proposed an AI-driven energy optimization system to cut consumption and emissions across production facilities.

CONCEPTS & TOOLS MASTERED

- Industrial IoT sensor data pipelines
- Predictive maintenance models
- Big-data infrastructure & real-time analytics

MEASURED IMPACT

20–30%

lower maintenance costs

50%

reduction in downtime

Real-time

anomaly detection

Where I can apply this work

From hospital floors to oil rigs, from city traffic to factory lines — the same AI workflow patterns translate.



Healthcare

Sensor-verified procedure compliance, predictive triage, hand-hygiene auditing.



Oil, Gas & Energy

Field-task verification, predictive maintenance, energy optimization.



Government & Public

Smart governance platforms, citizen engagement, fraud detection.



Smart Cities

Traffic flow, V2X coordination, digital-twin policy simulation.



Transportation

Autonomous coordination, fleet optimization, safety verification.



Agriculture

Precision farming, yield forecasting, crop-monitoring robotics.

LET'S CONNECT

Thank you.

Questions, opportunities, collaborations — I'm open to all three.



G I T H U B

<https://github.com/1plat4m/ITAI-2372-Artificial-Intelligence-Applications>



E M A I L

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A D D I T I O N A L S K I L L S

Python · HTML · CSS · Data Visualization · Applied AI Ethics